

Finiteness of the integer solutions of

$$x^4 + xy + y^3 + 1 = 0$$

Answer. The proposed infinitude statement is false. In fact, this equation has only finitely many rational solutions, and hence only finitely many integer solutions.

Scope of the proof. The algebraic calculations below are supplied in full. The finiteness conclusion uses Faltings' theorem, and the genus calculation uses the standard genus formula for a smooth plane curve. These results are stated explicitly and are not proved here. Thus the argument is unconditional, but it is not a fully self-contained elementary proof. A proof of the requested infinitude cannot be supplied because that conclusion is false.

Theorem 1. *The set*

$$\{(x, y) \in \mathbb{Q}^2 : x^4 + xy + y^3 + 1 = 0\}$$

is finite. It contains $(0, -1)$ and $(1, -1)$.

1. Intuition: check the curve before seeking a family

One equation in two variables need not have infinitely many integer solutions. Here the natural object to examine is the plane curve itself. The total degree is four, but that alone does not determine its arithmetic: factorization or singularities could change the situation substantially. We therefore check two precise facts: the curve is geometrically irreducible, and its projective completion is smooth. A smooth plane quartic has genus three, which places it within the scope of a theorem asserting finiteness even for rational points.

The most useful algebraic observation is that, at a hypothetical singular point, the derivative equations become linear relations among the three monomials x^4 , xy , and y^3 . These relations force incompatible absolute values. This makes the essential smoothness check particularly short.

2. Geometric irreducibility

Put $f(x, y) = x^4 + xy + y^3 + 1$. As a polynomial in y over the field $\mathbb{C}(x)$, it has degree three. A reducible cubic over a field has a root in that field. Suppose such a root were $r = p/q$, where $p, q \in \mathbb{C}[x]$ are coprime and $q \neq 0$. Clearing denominators gives

$$p^3 + xpq^2 + (x^4 + 1)q^3 = 0.$$

Thus q divides p^3 . Coprimality forces q to be constant, so r is a polynomial. If $\deg r \leq 1$, the term x^4 has strictly greater degree than every other term of $r^3 + xr + x^4 + 1$. If $\deg r = d \geq 2$, the term r^3 has degree $3d > \max\{d + 1, 4\}$ and cannot cancel. The zero polynomial is not a root either. This is a contradiction in every case.

Consequently f is irreducible over $\mathbb{C}(x)$ and over $\mathbb{C}[x, y]$: a factor depending only on x would have to divide the leading coefficient 1 in y , and any other factorization would persist over $\mathbb{C}(x)$.

3. The projective curve is smooth

The projective completion is

$$C : F(X, Y, Z) = X^4 + XYZ^2 + Y^3Z + Z^4 = 0 \quad \text{in } \mathbb{P}^2.$$

Its defining polynomial is irreducible over $\mathbb{C}$: a projective factorization would dehomogenize to one of f , unless a factor were a power of Z , and Z does not divide F .

Affine points. A singular point with $Z \neq 0$ would give complex numbers x, y satisfying

$$f(x, y) = 0, \quad 4x^3 + y = 0, \quad x + 3y^2 = 0.$$

Multiply the last two equations by x and y , respectively. We obtain

$$4x^4 + xy = 0, \quad xy + 3y^3 = 0.$$

Together with $x^4 + xy + y^3 + 1 = 0$, these force

$$\boxed{x^4 = \frac{3}{5}, \quad y^3 = \frac{4}{5}, \quad xy = -\frac{12}{5}.}$$

The first two equalities imply $|x| < 1$ and $|y| < 1$, even when x, y are complex. Hence $|xy| < 1$. The third equality instead gives $|xy| = 12/5 > 1$. Thus there are no affine singularities.

The point at infinity. Setting $Z = 0$ gives $X^4 = 0$, so the only point at infinity is $P = [0 : 1 : 0]$. Since

$$F_Z = 2XYZ + Y^3 + 4Z^3, \quad F_Z(P) = 1,$$

this point is nonsingular as well. Therefore C is smooth everywhere.

4. Genus and the finiteness conclusion

The genus formula for a smooth geometrically irreducible projective plane curve of degree d over a field of characteristic zero is

$$g = \frac{(d-1)(d-2)}{2}.$$

All its hypotheses were checked above, so our quartic has $g = 3$.

Faltings' theorem (used without proof). *If a smooth, projective, geometrically irreducible curve defined over a number field has genus at least two, then it has only finitely many points rational over that number field.*

Apply this theorem to C over $\mathbb{Q}$. Its rational point set $C(\mathbb{Q})$ is finite. Each rational solution of the original equation gives the distinct point $[x : y : 1] \in C(\mathbb{Q})$. Hence there are finitely many rational solutions, and therefore finitely many integer solutions. Finally,

$$0^4 + 0(-1) + (-1)^3 + 1 = 0, \quad 1^4 + 1(-1) + (-1)^3 + 1 = 0,$$

which verifies the two displayed solutions and completes the proof. $\square$

What is established. Infinitely many integer solutions are ruled out unconditionally. This argument supplies neither a height bound nor a complete list of integer solutions; in particular, it does not claim that $(0, -1)$ and $(1, -1)$ are the only ones. Faltings' theorem and the plane-curve genus formula are the explicitly identified external mathematical inputs.